

FAILSAFE: AI-Powered Student Failure Prediction and Early Intervention System

1. Introduction

FAILSAFE is an AI-powered academic early warning system developed to identify students who are at risk of academic failure before final examination results are released. The system combines Machine Learning, Explainable AI (SHAP), and an automated intervention recommendation engine to help faculty proactively support struggling students.

Unlike traditional prediction systems that provide only a risk score, FAILSAFE focuses on transparency and actionability by explaining why a student is at risk and recommending targeted interventions that can improve academic outcomes.

2. Dataset and Problem Definition

The project utilizes the UCI Student Performance Dataset, which contains academic, behavioural, demographic, and support-related information about students. To increase dataset size and improve model generalization, the Mathematics and Portuguese student datasets were concatenated into a single dataset.

The final grade (**G3**) was used to define the target variable.

Risk Classification

- **At Risk (1):** $G3 < 10$
- **Not At Risk (0):** $G3 \geq 10$

Although G3 was used to generate the target variable, it was excluded from the input features. Instead, the model learns to predict failure risk using earlier indicators such as G1, G2, attendance, study habits, lifestyle factors, and support systems.

3. Data Preprocessing and Model Development

The dataset underwent several preprocessing steps before training:

- Concatenation of Mathematics and Portuguese datasets.
- One-Hot Encoding of categorical features such as family support, internet access, higher education aspirations, and parental information.
- Generation of a binary target variable based on G3.
- Feature preparation and transformation.
- Train-test split using an 80:20 ratio.

An **XGBoost Classifier** was selected due to its strong performance on structured tabular datasets and its ability to capture complex relationships between academic, behavioural, and demographic factors.

Several hyperparameters were manually tuned to improve model performance and generalization before selecting the final model configuration.

4. Model Evaluation

The final model achieved strong predictive performance on the test dataset.

MODEL PERFORMANCE		CONFUSION MATRIX		
Accuracy	93.81%		Predicted safe	Predicted at risk
Precision	89%	Actually safe	158	5
Recall	83%	Actually at risk	8	39
F1 Score	86%			

The results indicate that the model effectively identifies students who require academic intervention while maintaining a low false-positive rate. This balance is important in educational environments where unnecessary interventions should be minimized while still identifying vulnerable students.

5. Explainable AI Framework

A key objective of FAILSAFE was to ensure that predictions remain interpretable and trustworthy for faculty members.

After a prediction is generated, the processed student data is passed to a SHAP explainer. SHAP calculates the contribution of each feature toward the predicted failure probability.

However, displaying every SHAP contribution can overwhelm users and may highlight factors that are not educationally meaningful. To address this, FAILSAFE adopts a hybrid explainability framework that combines SHAP values with predefined educational risk rules.

Examples include:

Feature	Risk Condition
G1	< 10
G2	< 10
Absences	≥ 8
Study Time	≤ 2
Previous Failures	> 0
Travel Time	≥ 3
Weekend Alcohol Consumption	≥ 4
Weekday Alcohol Consumption	≥ 3
Family Support	Not Available
Higher Education Goal	Not Present

A feature is considered a significant risk factor only when:

1. The predefined educational threshold is satisfied.
2. The SHAP value positively contributes to failure risk.

The resulting factors are ranked according to SHAP impact, and the **Top 5 Risk Factors** are selected and presented to the faculty member. This ensures that explanations remain both data-driven and actionable.

6. Intervention Recommendation Engine

The identified risk factors are passed to an Intervention Engine that converts predictive insights into actionable recommendations.

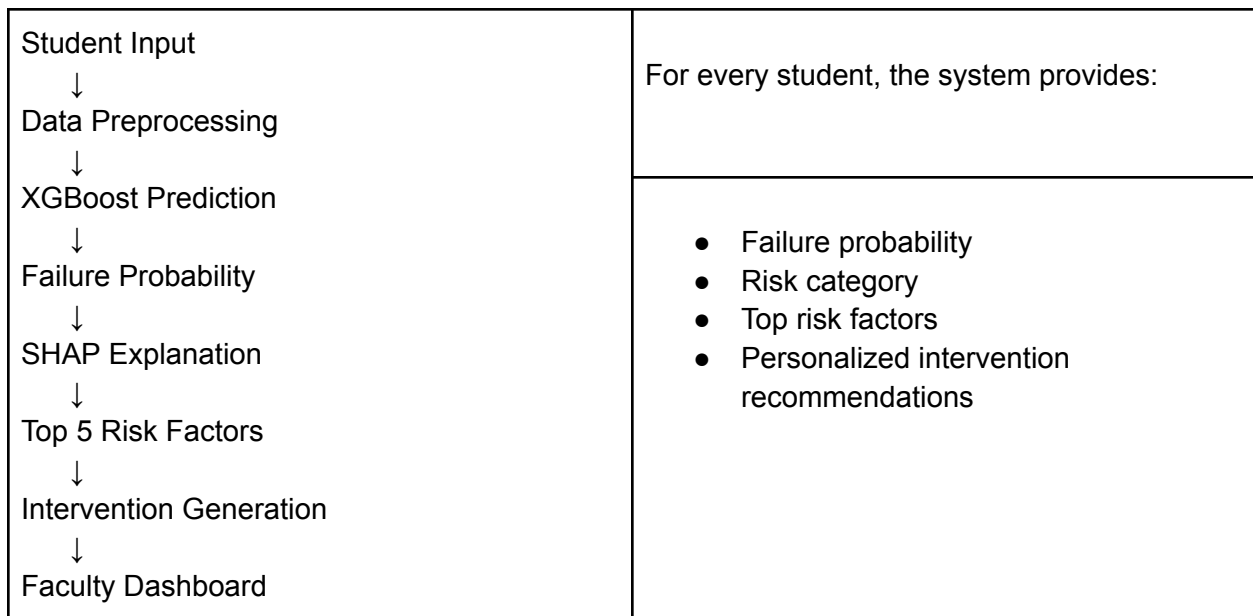
Each risk factor is mapped to a corresponding educational intervention strategy. Examples include:

- Low academic performance → Remedial support and tutoring.
- High absenteeism → Attendance counselling.
- Previous failures → Academic mentoring.
- Low study time → Structured study plans.
- Weak family support → Family engagement initiatives.
- High alcohol consumption → Wellness counselling.

Each recommendation includes the intervention title, recommendation description, associated risk factor, and severity level. This enables faculty members to move beyond prediction and immediately implement targeted support measures.

7. System Workflow

The overall workflow of FAILSAFE is illustrated below:



8. Class Analysis Module

To extend the system beyond individual predictions, a Class Analysis module was developed.

Faculty members can upload a CSV file containing multiple student records. The system automatically processes each student using the same prediction and explainability pipeline.

Students are categorized into:

- Low Risk
- Medium Risk
- High Risk-Highlighted on the Dashboard

High-risk students are highlighted to enable rapid identification and intervention.

The dashboard also provides class-level insights including:

- Total number of students
- Number of low-risk students
- Number of medium-risk students
- Number of high-risk students
- Risk distribution visualization

Faculty can additionally select any student within the uploaded class to view their detailed risk probability, top risk factors, and personalized interventions.

9. Conclusion

FAILSAFE successfully integrates Machine Learning, Explainable AI, and educational intervention planning into a unified platform. By combining accurate risk prediction, interpretable explanations, and automated intervention recommendations, the system enables educators to identify struggling students early and take proactive measures before academic failure occurs.

The project demonstrates how AI can be applied responsibly within education to support data-driven decision-making and improve student success.